

UNIT 7

FLUID DYNAMICS



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FLUID DYNAMICS:

"The study of fluid behavior in motion and its applications is called fluid dynamics."

FLUID FRICTION:

"The force that opposes a motion that takes place within fluids or in between fluids is called fluid friction."

Fluid friction is categorized into two types:

- Viscous force
- Drag force

VISCOSITY:

"Viscosity is the property of the fluid by virtue of which it opposes the relative motion between its adjacent layers."

VISCOUS FORCE:

"The force of friction between the layers of the fluid, which opposes the flow of fluid is called viscosity."

EXPRESSION FOR VISCOUS FORCE OR NEWTON'S LAW OF VISCOSITY:

For a given fluid, it is required force F which is proportional to the area of fluid in contact with each layer 'A', and to the speed, 'v', and is inversely proportional to the separation 'L', between the layers,

$$F \propto v \times \frac{A}{L}$$

$$F = \eta \times \frac{A}{L} \times v$$

Where η is known as the coefficient of viscosity which is an intrinsic property of fluid.

COEFFICIENT OF VISCOSITY:

"The ratio of the shearing stress to the velocity gradient of the fluid is called the coefficient of viscosity η ."

$$\eta = \frac{F}{A} \times \frac{1}{v/L}$$

Unit of coefficient of friction:

The SI unit for coefficient of friction is $\text{N.s/m}^2 = \text{Pa.s}$ (Pascal Second).

STOKE'S LAW:

If a spherical body of radius r falls in a fluid of viscosity η with a velocity v , the drag force or fluid friction act upon it, is:

$$F = 6\pi\eta rv$$

TERMINAL VELOCITY:

"When the weight of the falling body and drag force become equal in magnitude, the velocity of the body becomes maximum (constant) which is called terminal velocity."

Expression for terminal velocity:

Consider a sphere starts its motion from rest i.e., $v=0$, therefore $a=g$ in the beginning. But as the velocity of the sphere increases until it just balances the weight of the body.

According to Stokes' law, the drag force acting on the sphere is,

$$F = 6\pi\eta rv$$

The gravitational force given by,

$$w = mg$$

$$W = \rho v g$$

$$W = \frac{4}{3} \pi r^3 \rho g \dots (2)$$



When sphere is moving downward, an upthrust is also acting,

$$F = \frac{4}{3}\pi r^3 \rho g \dots (3)$$

When sphere attains terminal velocity, drag force and upthrust balance the gravitational force,

$$F + F_u = W$$

$$\frac{4}{3}\pi r^3 \rho g + 6\pi \eta r v = \frac{4}{3}\pi r^3 \rho g$$

$$6\pi \eta r v = \frac{4}{3}\pi r^3 \rho g$$

$$v_r = \frac{4r^2 g(\sigma - \rho)}{9 \times \eta}$$

$$v_r = \frac{2r^2 g(\sigma - \rho)}{9 \times \eta}$$

TYPES OF FLUID FLOW:

• Steady flow:

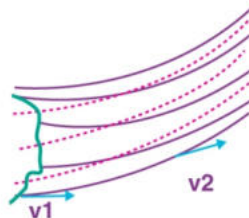
"The flow in which the fluid characteristics like velocity, pressure and density at a point do not change with time then it is called steady flow."

• Unsteady flow:

"The flow in which the velocity, pressure and density change with time then it is called unsteady flow."

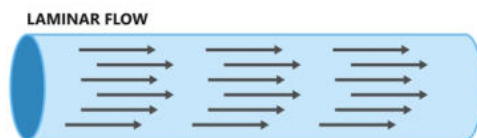
• Streamline flow:

"The flow in which each particle of the liquid passing through a point travel along the same path and with the same velocity is called streamline flow." [Stream lines are straight or curve path of steady fluid]



• Laminar flow:

"It is the fluid flow in which particles move along well-defined stream lines which are straight and parallel. In laminar flow the velocities at different points may have different magnitudes but their directions are parallel."



• **Turbulent flow:** "The irregular flow in which the particles can move in different directions due to which eddies formation take place which are responsible for high energy loss is called turbulent flow."



• Incompressible fluid flow:

"Incompressible fluid flow is a type of flow in which the density of the fluid remains constant as it moves through regions of varying pressure. This means that the fluid does not change its volume or area."

• Non-viscous fluid flow or ideal fluid (Inviscid flow):

"An incompressible, non-viscous fluid is called an ideal fluid with zero viscosity flows and no dissipation of energy."

CRITICAL VELOCITY:

"The velocity of liquid flow up to which its flows are streamline and above which its flows become turbulent is called critical velocity."

REYNOLD'S NUMBER:

"A pure number which determines the nature of flow of liquid is called Reynolds number."

$$Re = \frac{\rho v L}{\eta}$$

$$Re = \frac{\rho v d}{\eta}$$

Where,

ρ = Density of the fluid

V = Mean speed of fluid

d = Diameter of tube

η = Coefficient of viscosity

Reynold's number condition for fluid flow:

• If $0 < Re < 2300$, then flow is streamline or laminar.

• If $2300 < Re < 4000$, then flow is transient.

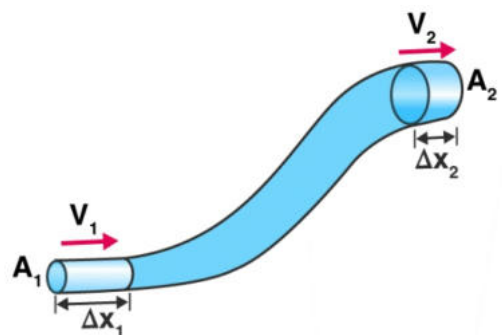
• If $Re > 4000$, then flow is turbulent.

EQUATION OF CONTINUITY: The continuity equation is the mathematical expression of the law of conservation of mass in fluid dynamics.

"In incompressible, steady fluid flow, mass of fluid entering into the tube in a particular time interval is equal to the mass of fluid leaving the tube in that particular time."

$$\rho_1 A_1 \Delta x_1 = \rho_2 A_2 \Delta x_2$$

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$



$$A_1 V_1 = A_2 V_2$$

Hence, volume of water flowing per second (Volume flow rate) remains constant.

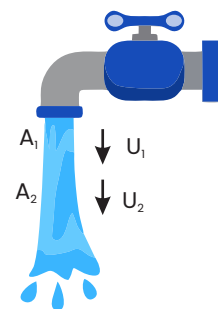
$$AV = \text{Constant}$$

The velocity of the flow will increase if the cross-section decreases and vice versa.

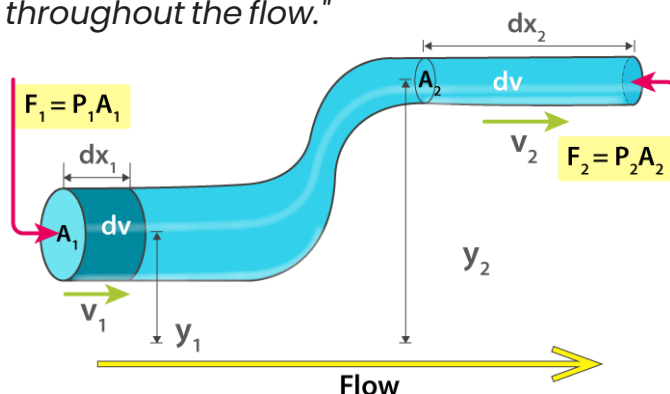
Example:

When water flows from a tap, the velocity of falling water under the action of gravity will increase with distance from tap i.e., $V_2 > V_1$.

So, in accordance with the continuity equation the cross section of the water stream will decrease $A_1 < A_2$ and the falling stream of water becomes narrower.

**BERNOULLI'S THEOREM:**

"The total mechanical energy (Pressure energy, Potential energy & kinetic energy) per unit volume or per unit mass of incompressible and non-viscous fluid in steady flow through a pipe remains constant throughout the flow."

**Mathematically:**

Consider a pipe whose diameter and elevation changes as fluid passes through it. Consider small mass of the fluid with density ρ that flows from point 1 to 2 as shown in figure.

Work done:

The work done by a force F on the fluid to displace it by an infinitesimal distance Δx is given by,

$$W = F\Delta x$$

Therefore, at point 1 and point 2 works done are,

$$W_1 = F_1\Delta x_1$$

$$W_2 = F_2\Delta x_2$$

Total work done when fluid moves from point 1 to 2 is,

$$\Delta W = W_1 - W_2$$

$$\Delta W = F_1\Delta x_1 - F_2\Delta x_2$$

$$\Delta W = P_1A_1\Delta x_1 - P_2A_2\Delta x_2$$

$$\Delta W = P\Delta V \text{ (1)}$$

Kinetic Energy:

Now, when the fluid moves from point 1 to 2 there is a change in kinetic energy,

$$\Delta W = p_1A_1\Delta x_1 - p_2A_2\Delta x_2 = p_1\Delta V - p_2\Delta V = (p_1 - p_2)\Delta V$$

Now, when the fluid moves from point 1 to 2, there is a change of kinetic energy.

$$\Delta K.E = \frac{1}{2} m_2 v_2^2 - \frac{1}{2} m_1 v_1^2 = \frac{1}{2} \rho \Delta V v_2^2 - \frac{1}{2} \rho \Delta V v_1^2 = \frac{1}{2} \rho \Delta V (v_2^2 - v_1^2)$$

Similarly, the change in potential energy is given by,

$$\Delta U = mgh_2 - mgh_1 = \rho \Delta V g (h_2 - h_1)$$

The work done in moving the fluid is the sum of the change in kinetic and potential energies.

$$\Delta W = \Delta K.E + \Delta U$$

$$\text{or, } (p_1 - p_2)\Delta V = \frac{1}{2} \rho \Delta V (v_2^2 - v_1^2) + \rho \Delta V g (h_2 - h_1)$$

$$\text{or, } p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g (h_2 - h_1)$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$\text{or, } p + \frac{1}{2} \rho v^2 + \rho g h = \text{constant}$$

This is Bernoulli's equation.

Therefore

$$P + \frac{1}{2} \rho v^2 + \rho g h = \text{Constant} \dots$$

The height h is measured from some convenient reference level, for example at the surface of the liquid. The equation is really a statement of conservation of energy per unit volume in the fluid.

- $\frac{1}{2} \rho v^2 + \rho g h$ is total pressure.
- $\rho g h$ is static pressure.
- $\frac{1}{2} \rho v^2$ is dynamic pressure.

APPLICATIONS OF BERNOULLI'S THEOREM:

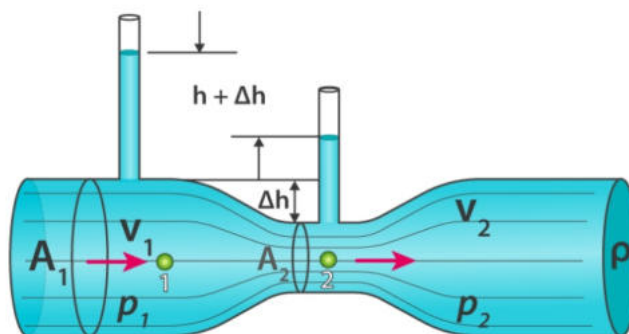
• Filter Pump:

In a filter pump, Bernoulli's principle is applied to increase the pressure of the fluid passing through the pumping. By reducing the cross-sectional area of the pump's outlet, the fluid's velocity increases according to the principle, leading to a decrease in pressure (as kinetic energy increases). This pressure drop helps draw fluid into the pump and through the filter medium, facilitating the filtration process.



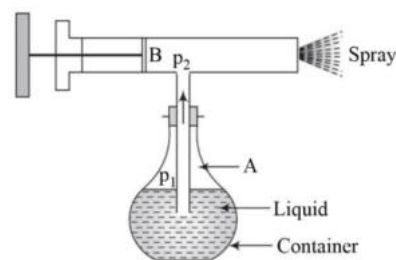
• Venturi Meter:

A Venturi meter is a device used to measure the flow rate of a fluid in a pipe. It consists of a gradually narrowing tube (Venturi tube) inserted in the pipe. As the fluid flows through the narrowing section, its velocity increases according to Bernoulli's principle, and the pressure decreases. By measuring the pressure difference between the narrowest section and the wider parts of the pipe, the flow rate can be determined.



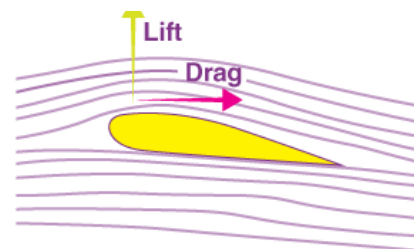
• Atomizer:

Atomizers are devices used to convert liquids into fine sprays or mists. The application of Bernoulli's principle is crucial in this process. As the liquid passes through a small nozzle in the atomizer, its velocity increases, leading to a decrease in pressure according to the principle. This decrease in pressure facilitates the breakup of the liquid into tiny droplets or a fine spray.



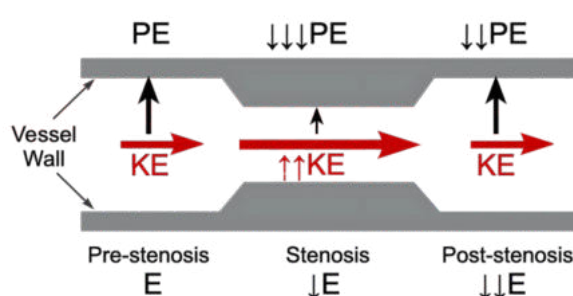
• Flow of Air over an Aero foil:

When air flows over an aero foil (such as the wing of an aircraft), the shape of the aero foil causes the air to travel faster over the top surface compared to the bottom surface. According to Bernoulli's principle, the air pressure decreases over the top surface due to the increased velocity, creating a pressure difference that results in lift, allowing the aircraft to stay airborne.



• Blood Physics:

In the circulatory system, Bernoulli's principle is applicable to the flow of blood through blood vessels, particularly in areas of constriction or stenosis. When the diameter of a blood vessel narrows, the blood velocity increases, leading to a decrease in pressure, which can have clinical implications in conditions such as atherosclerosis or stenosis. It's essential to note that while Bernoulli's principle is an excellent theoretical tool for understanding fluid behavior in these applications, real-world fluids may have additional complexities, such as viscosity and compressibility, which need to be considered for precise calculations and analyses.



PAST TEN YEARS QUESTIONS

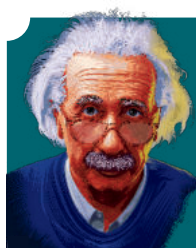
Bernoulli's Principle:

Q1: What is Bernoulli's principle? Derive Bernoulli's equation for steady, incompressible, non viscous, and irrotational flow of fluid. (2024)

Numerical:

Q2: Calculate the viscous drag on a drop of oil of 0.1 mm radius falling through air at its terminal velocity.

(Viscosity of air = $1.8 \times 10^{-5} \text{ Pa.s}$, Density of oil 850 Kg/m^3) (2024)



Once you stop learning
you start dying

ALBERT EINSTEIN

UNIT 8

ELECTRIC FIELD

